Project 2: Decomposition of monthly souvenir sales

Trend, seasonality and multiplicative model selection

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Context

The dataset `data/SouvenirSales.csv` contains the monthly sales (in Australian dollars) of a souvenir shop on the beach of Queensland (Australia), from January 1995 to December 2001. The series exhibits both a strong upward trend and a marked seasonal pattern with a Christmas spike — a textbook example of a *multiplicative* time series.

Deliverables

- A reproducible Python script (`project2_<name>.py`, seed 42).
- A 4–6 page PDF report with figures, tables and discussion.
- Optional 5-minute oral defence.

Exercise 1 – Exploratory analysis

(1) Load the data, parse the dates, plot Y_t vs. time.
(2) Comment on:

- the trend (is it roughly linear, exponential, or something else?);
- the seasonal amplitude (constant or growing with the level?);
- the presence of any obvious outliers.

(3) Compute and report descriptive statistics (mean, std, min, max, skewness, kurtosis).

Exercise 2 – Additive vs. multiplicative

(1) Apply an additive decomposition with `statsmodels.tsa.seasonal_decompose(y, period=12, model="additive")` and plot the four panels.
(2) Same with `model="multiplicative"`.
(3) Compare the residuals of the two models. Which is more homoscedastic?
(4) Conclude: which model is more appropriate for this series, and why?

Exercise 3 – Log transform and additive workflow

(1) Take $\log Y_t$. Plot the log-series.
(2) On $\log Y_t$, apply the full additive workflow:

- estimate the trend (LOESS, frac $= 0.3$, or polynomial degree 2);
- detrend;

- estimate seasonality with seasonal averaging *and* a cosine–sine model with one harmonic;
- compute the residual.

(3) Diagnose the residual: histogram, QQ-plot, time-vs-residual plot. Is it stationary?

Exercise 4 – Forecasting

(1) Split the data: train = 1995-01 to 2000-12 (72 months), test = 2001-01 to 2001-12 (12 months).
(2) On the train set, fit a model $\log Y_t = \beta_0 + \beta_1 t + \sum_{j=1}^{2}[a_j \cos(j\omega t) + b_j \sin(j\omega t)] + \varepsilon_t$ ($\omega = 2\pi/12$). Use OLS.
(3) Predict $\widehat{\log Y_t}$ for t in the test range. Convert back: $\widehat{Y}_t = \exp(\widehat{\log Y_t})$.
(4) Compute the test MSE and MAPE. Plot observed vs. predicted on the test period.

Exercise 5 – Bonus: residual modelling

(1) Plot the autocorrelation function of the train residual (`statsmodels.graphics.tsaplots.plot_acf`).
(2) Are there significant autocorrelations beyond lag 0? This would suggest that the residual is not white noise — and motivates the ARMA modelling of Chapter 5.

Marking grid (out of 100)

Criterion	Points
Exploratory plots + descriptive statistics	15
Additive vs. multiplicative comparison + discussion	20
Log + full additive workflow + residual diagnostic	25
Forecast on test set: train/test split, OLS, MAPE	20
ACF of residual + ARMA motivation (bonus)	10
Reproducibility + report quality	10
Total	**100**

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